



Institut Universitaire des sciences

- Faculté des sciences et de technologie

✓ TD N°7 Réseau II

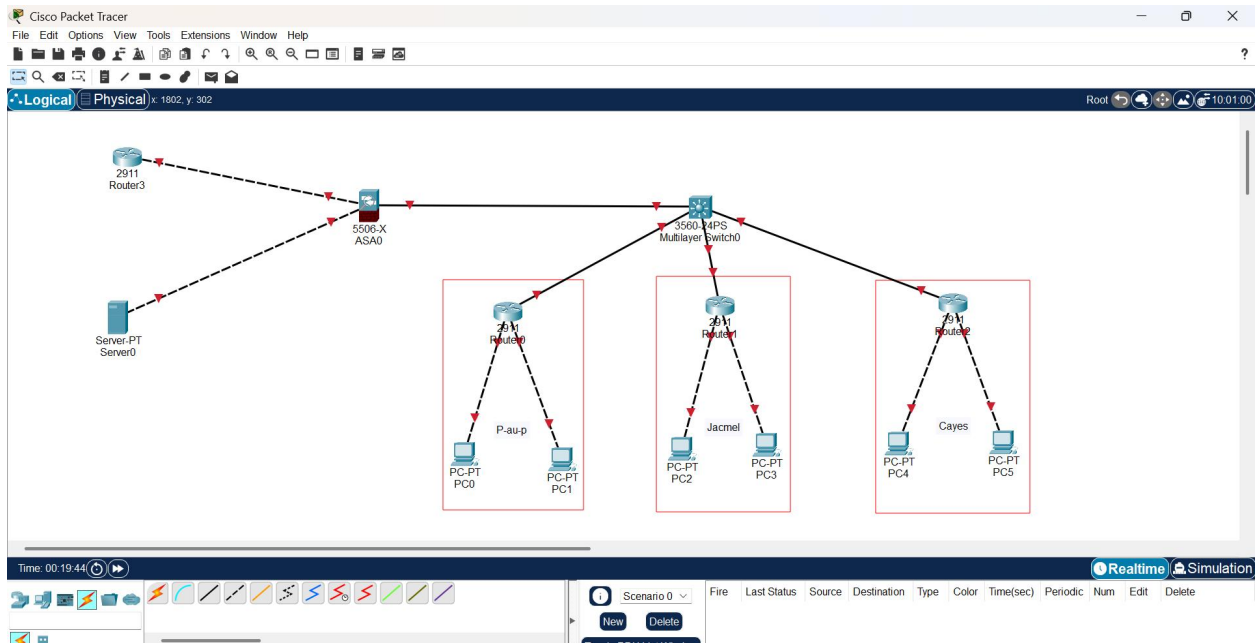
Nom & Prénom : COFFY Cliford

Niveau : L3

Prof. : Ismaël SAINT AMOUR

Date : 09/05/2026

Creation du Topologie



Configuration des Routers

Router P-au-p

Tracer 8.2.2\save\TD7Reseau11.pkt

Help

5506-X ASA0

PC-PT PC0

Router0

IOS Command Line Interface

```
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no sh

% Invalid input detected at '^' marker.

Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#
```

Copy Paste

Simulation

delete

tomatically Choose Connection Type

Router Jacmel

Router1

Physical Config CLI Attributes

IOS Command Line Interface

DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip address 192.168.20.1 255.255.255.0

% Invalid input detected at '^' marker.

Router(config)#interface g0/0

Router(config-if)#ip address 192.168.20.1 255.255.255.0

Router(config-if)#no sh

Router(config-if)#

%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1

Router(config-if)#ip address 10.0.0.6 255.255.255.252

Router(config-if)#no sh

Router(config-if)#

%LINK-S-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router ospf 1

Router(config-router)#network 192.168.20.0 0.0.0.255 area 0

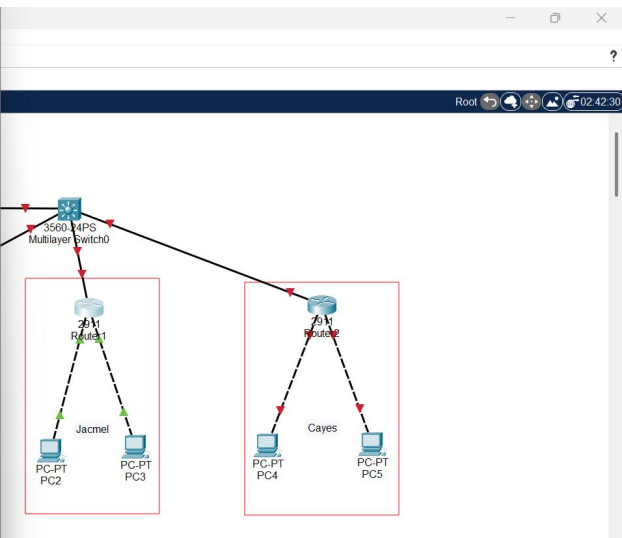
Router(config-router)#network 10.0.0.4 0.0.0.3 area 0

Router(config-router)#

Copy Paste

☐ Top

Root 02:42:30



Realtime Simulation

Scenario 0 Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

Delete

U List Window

Router Cayes

Router2

Physical Config CLI Attributes

IOS Command Line Interface

Cisco C1802S11/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400K8
4 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface g0/0

Router(config-if)#ip address 192.168.30.1 255.255.255.0

Router(config-if)#no sh

Router(config-if)#

%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1

Router(config-if)#ip address 10.0.0.10 255.255.255.252

Router(config-if)#no sh

Router(config-if)#

%LINK-S-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router ospf 1

Router(config-router)#network 192.168.30.0 0.0.0.255 area 0

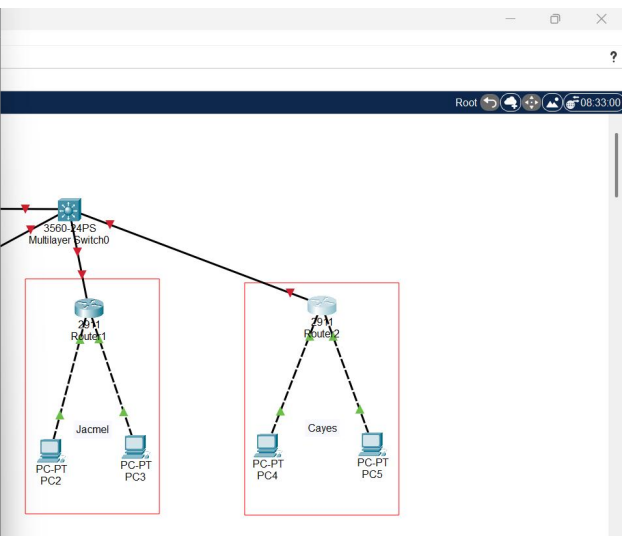
Router(config-router)#network 10.0.0.8 0.0.0.3 area 0

Router(config-router)#

Copy Paste

☐ Top

Root 08:33:00



Realtime Simulation

Scenario 0 Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

Delete

U List Window

Address ip aux PC

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::240:BFF:FE1D:744

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC1

Physical Config Desktop Programming Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address192.168.10.3

Subnet Mask255.255.255.0

Default Gateway192.168.10.1

DNS Server0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local AddressFE80::20A:F3FF:FE53:5456

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

AuthenticationMD5

Username

Password

☐ Top

es\TD7ReseauII. PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.30.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.30.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20C:85FF:FE37:C442

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Connection Type

alttime Edit

Address IP au Server

Server WEB

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.100.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.100.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::290:2BFF:FE9C:A6A0

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Configuration du Firewall

The screenshot shows the Cisco Packet Tracer interface with the ASA0 configuration window open. The network diagram on the left shows a 2911 Router3 connected to a 5506-X ASA0, which is also connected to a Server-PT (Server WEB). The ASA0 configuration window displays the following commands:

```
INFO: Power-On Self-Test complete.
INFO: Starting HW-DRBG health test...
INFO: HW-DRBG health test passed.
INFO: Starting SW-DRBG health test...
INFO: SW-DRBG health test passed.

Type help or '?' for a list of available commands.

ciscoasa>
ciscoasa>
ciscoasa>en
ciscoasa#conf t
ciscoasa(config)#int gi/1
ciscoasa(config-if)#nameif outside
INFO: Security level for "outside" set to 0 by default.
ciscoasa(config-if)#security-level 0
ciscoasa(config-if)#ip address 203.0.113.2 255.255.255.0
ciscoasa(config-if)#no sh

ciscoasa(config-if)#
ciscoasa(config-if)#int gi/3
ciscoasa(config-if)#nameif dmz
INFO: Security level for "dmz" set to 0 by default.
ciscoasa(config-if)#security-level 50
ciscoasa(config-if)#ip address 192.168.100.1 255.255.255.0
ciscoasa(config-if)#no sh

%LINK-5-CHANGED: Interface GigabitEthernet1/3, changed state to down
ciscoasa(config-if)#int gi/2
ciscoasa(config-if)#nameif inside
INFO: Security level for "inside" set to 100 by default.
ciscoasa(config-if)#security-level 100
ciscoasa(config-if)#ip address 10.0.0.1 255.255.255.252
ciscoasa(config-if)#no sh
```

Configuration du router Internet

The screenshot shows the Cisco Packet Tracer interface with the Router3 configuration window open. The network diagram on the left shows a 2911 Router3 connected to a 5506-X ASA0, which is also connected to a PC-PT (PC). The Router3 configuration window displays the following commands:

```
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX1524008S
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249956K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#conf t
Router(config)#
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#no sh

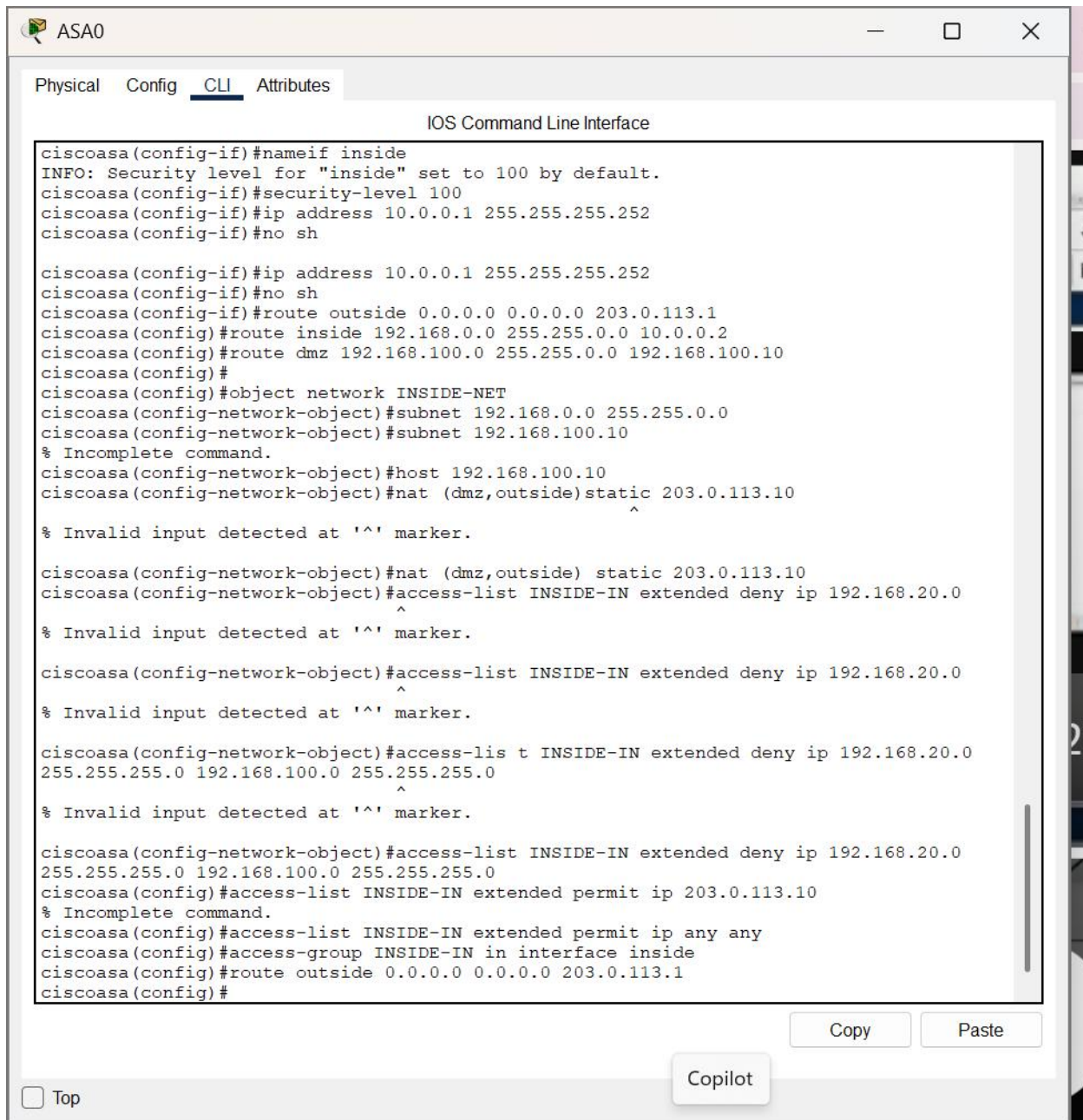
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#int g0/1
Router(config-if)#ip address 192.51.100.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#ip route 0.0.0.0 0.0.0.0 192.51.100.2
Router(config)#ip route 192.168.0.0 255.255.0.0 203.0.113.2
Router(config)#
```

Fin du configuration du firewall



The screenshot shows a terminal window titled "ASA0" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The configuration commands and their outputs are as follows:

```
ciscoasa(config-if)#nameif inside
INFO: Security level for "inside" set to 100 by default.
ciscoasa(config-if)#security-level 100
ciscoasa(config-if)#ip address 10.0.0.1 255.255.255.252
ciscoasa(config-if)#no sh

ciscoasa(config-if)#ip address 10.0.0.1 255.255.255.252
ciscoasa(config-if)#no sh
ciscoasa(config-if)#route outside 0.0.0.0 0.0.0.0 203.0.113.1
ciscoasa(config)#route inside 192.168.0.0 255.255.0.0 10.0.0.2
ciscoasa(config)#route dmz 192.168.100.0 255.255.0.0 192.168.100.10
ciscoasa(config)#
ciscoasa(config)#object network INSIDE-NET
ciscoasa(config-network-object)#subnet 192.168.0.0 255.255.0.0
ciscoasa(config-network-object)#subnet 192.168.100.10
% Incomplete command.
ciscoasa(config-network-object)#host 192.168.100.10
ciscoasa(config-network-object)#nat (dmz,outside)static 203.0.113.10
^
% Invalid input detected at '^' marker.

ciscoasa(config-network-object)#nat (dmz,outside) static 203.0.113.10
ciscoasa(config-network-object)#access-list INSIDE-IN extended deny ip 192.168.20.0
^
% Invalid input detected at '^' marker.

ciscoasa(config-network-object)#access-list INSIDE-IN extended deny ip 192.168.20.0
^
% Invalid input detected at '^' marker.

ciscoasa(config-network-object)#access-list INSIDE-IN extended deny ip 192.168.20.0
255.255.255.0 192.168.100.0 255.255.255.0
^
% Invalid input detected at '^' marker.

ciscoasa(config-network-object)#access-list INSIDE-IN extended deny ip 192.168.20.0
255.255.255.0 192.168.100.0 255.255.255.0
ciscoasa(config)#access-list INSIDE-IN extended permit ip 203.0.113.10
% Incomplete command.
ciscoasa(config)#access-list INSIDE-IN extended permit ip any any
ciscoasa(config)#access-group INSIDE-IN in interface inside
ciscoasa(config)#route outside 0.0.0.0 0.0.0.0 203.0.113.1
ciscoasa(config)#
```

At the bottom of the window, there are buttons for "Copy", "Paste", and "Copilot", along with a "Top" link.

Test Ping

The network diagram shows a topology with three devices: Router3 (2911), ASA0 (5506-X), and Server-PT (Server WEB). Router3 is connected to ASA0, and ASA0 is connected to Server-PT. The PC0 command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
c:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

c:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=6ms TTL=128
Reply from 192.168.10.2: bytes=32 time=9ms TTL=128
Reply from 192.168.10.2: bytes=32 time=7ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 9ms, Average = 6ms

c:\>
```

The network diagram is identical to the one above. The PC0 command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

c:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=6ms TTL=128
Reply from 192.168.10.2: bytes=32 time=9ms TTL=128
Reply from 192.168.10.2: bytes=32 time=7ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 9ms, Average = 6ms

c:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

c:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

c:\>
```

Questions ? / Réponses

1. NAT statique vs dynamique

- **Statique** : une IP privée est toujours traduite vers la même IP publique (liaison fixe).
- **Dynamique** : une IP privée peut être traduite vers différentes IP publiques disponibles (liaison flexible).

2. ACL sur ASA

Une **ACL (Access Control List)** sur un firewall ASA filtre le trafic en autorisant ou bloquant les paquets selon des règles (IP source/destination, protocole, port).

3. Segmenter les réseaux par site

Cela permet de **limiter la propagation des problèmes**, d'améliorer la **sécurité**, et de faciliter la **gestion du trafic** entre différentes zones géographiques.

4. Clé pré-partagée

Dans un VPN, une clé pré-partagée est utilisée pour **authentifier les deux extrémités** avant d'établir le tunnel sécurisé.

5. ACL dans un VPN

Elle définit **quels flux sont autorisés à passer dans le tunnel VPN** (par exemple, seulement le trafic entre deux sous-réseaux précis).

6. Chiffrer communications inter-sites

Le chiffrement garantit la **confidentialité** et l'**intégrité** des données échangées entre sites, empêchant l'interception ou la modification par des tiers.

7. Phase 1 vs Phase 2

- **Phase 1 (IKE)** : négociation initiale, authentification et création du canal sécurisé.
- **Phase 2 (IPSec)** : établissement du tunnel de données et définition des paramètres de chiffrement pour le trafic réel